

Representation Adequacy and Transportable Relational Profiles

Omics studies measure the activity of thousands of genes or other molecules in each patient. A common aim is to discover groups of people who may share disease mechanisms. The resulting groups, however, depend strongly on how patients are compared. We can compare measurement values directly, or ask which genes are more active than others within the same patient. The second approach produces simple rules such as “gene A is more active than gene B”. These rules may be easier to interpret and less dependent on laboratory scale, but they are not always better.

This project will determine when each way of comparing patients is supported, and will then develop Transportable Relational Patient Profiles where the evidence justifies them. Using only one discovery cohort, the software will test comparisons based on values, ranks, gene-to-gene relations, or a combination of these views. If none provides credible structure, the software will be allowed to withhold clustering. For supported groups, it will learn short profiles made of relations between molecular features. Such a profile will provide a readable description of a group instead of an artificial “average patient”.

The complete profile will then be frozen and applied to patients from independent cohorts. New-patient data will not be used to retrain the method, normalise cohorts jointly, or improve thresholds. A patient will be assigned only when enough rules can be observed and the decision is sufficiently clear. Otherwise the patient will remain unassigned. Uncertainty will therefore be reported instead of being hidden by a forced decision.

The research will combine controlled simulations with independent lung and colorectal cancer cohorts. In every primary comparison, different patient representations will use the same deterministic clustering algorithm. We will measure profile stability, length and readability, gene coverage across platforms, resistance to perturbations, and agreement of frozen groups with biological and clinical information. Disease labels and clinical outcomes will be used only at the final evaluation stage.

Preliminary results show that different representations work best under different conditions. They also reveal an important limitation: stable data structure need not match a particular clinical label, and one of two transfer directions between cohorts failed a threshold fixed in advance. The project therefore does not assume that relations will always win. Its outcome will be a set of methods and transparent rules defining when relational patient profiles are credible, when measurement values should be retained, and when withholding clustering is the honest conclusion.